

Phase 4 Report

Brief Description of Game

Graphite Gambit is a top-down retro-styled 2D game where you are a pencil, trying to complete a connect-the-dots shape in order to unlock the exit and progress to the next level. However, achieving this goal is costly, and the price is graphite! Each step depletes some graphite (with each level making these steps more expensive!), so you must be strategic with your movement while continually collecting randomly-spawning graphite shards. Explore the map, consisting of random doodles and school supplies, while navigating around a marker maze. Try not to trip over white-out, which deals with large graphite loss, and avoid ink spills, which slow you down greatly. And while you're strolling, don't get caught by your two biggest enemies: the eraser and the pencil sharpener. Getting attacked by an eraser means large graphite loss. But getting caught by the pencil sharpener is even worse, as it holds you in place and depletes your graphite over time! If these steps are followed successfully, while falling under the 300-second time limit, a trophy will be waiting for you at the exit, where you can claim your victory.

Deviations from Original Plan and Justification

In our finished game, we had deviated in various parts from our original plan that we had made. Some changes include deviations from our UML and core game logic.

Initially, since we did not have a very good understanding of how our final product would look, our UML was left lacking many required functionalities that were needed later in the development of our game. This resulted in several flaws, such as high coupling from world to entity, and the world ended up as a god class that handled almost the entire gameplay logic, which was later refactored to have less responsibility with the separation of collision logic.

Additionally, we had various other plans for our game, but did not have a clear idea of how the gameplay would look. One such change in plans was with the eraser entity, as it was planned that they would just collide with the player and then die. However, due to the player having access to an endless supply of spawned graphite to heal, this was changed into a simple respawn feature where the erasers would revive after some time to keep the game difficult.

Another plan that was scrapped was within the pickup system itself. The original plan was to have two separate pickups aside from the plot points, which were graphite to heal and wood shavings, which would give points. Instead, we decided to remove the wood shavings from our plan and ended up awarding any excess healing from the graphite as points to the player, which helped to reward players who could remain at high health and move optimally.

Ultimately, the UML and initial plan were followed quite closely, but later into development, we had to make some changes to our design in order to implement new features, improve our gameplay experience, and improve the quality of our code.

Important Lessons Learned

During the development of the game, we learned several important lessons. First of all, we realized how important the initial design is. As more and more features were being added, and

deadlines started becoming more of a worry, it made us implement features without caring for the quality of the code. Rather, we had a “just get it done” mentality. An example of this is that most of us had code smells in our classes, especially duplication. We would trial and error certain features or bug fixes until something worked, and once that one idea worked, we would copy and paste it all over without a worry for maintainability. Unfortunately, this thinking caused us to have expensive fixes toward the later stages of our code, as we had to carry out a refactoring stage that addressed many of these problems. This taught us that it is more difficult to change what has already been done, so build it correctly from the beginning.

Also, we learned not to lose sight of the need for regular testing. While working on the game, we encountered bugs that initially seemed insignificant. Such errors include the logic of building a route from enemies to the player. The algorithm was not optimized when there were more enemies on the map; this caused the route to be frequently and unnecessarily recalculated, and thus impacted our performance. As a result, we had to redesign the algorithm such that its core behavior would not change; something that should have been addressed much earlier, but was not known about due to our lack of testing. This taught us an important lesson that testing should not be one of the final stages. Instead, it should be accompanied throughout the entire development process.

Finally, the most important lesson was team communication. From experience, we realized that bad communication can lead to many problems, such as a weak understanding or unfair assignment of duties. We learned the value of communicating with purpose, which means refraining from staying quiet in meetings or agreeing blindly even when you have your own opinions about something. It also means taking initiative when others wouldn't. For example, there have been many situations where communication within our group was poor, and a direct byproduct of this was a lack of performance among members. But when a few members stepped up and communicated clearly, this created a sense of direction and purpose that wasn't present before. The key takeaway is that effective communication is absolutely vital; when we refuse to settle for ambiguity, the group dynamic can switch from just passively participating to a more cohesive environment, one where we all share a sense of direction.

How to Play

Our game uses both WASD and arrow keys to control the pencil's movement. To progress from one level to the next, you must collect all the plot points to complete the connect-the-dots shape. This opens the end room's doors, which stay locked until this condition is met.

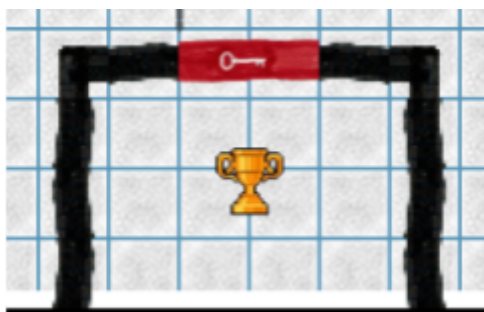


Image 1: Locked end room



Image 2: Connect the dots shape is created, so end room door opens

When the player collides with the trophy within the end room, the win screen is displayed. The option to continue to the next level is now unlocked, or the user can quit to the main menu.

Regardless of which option they choose, their progress will be saved to a Save directory, containing save.json, which records their levels unlocked and high scores for each of these levels.



Image 3: Win screen

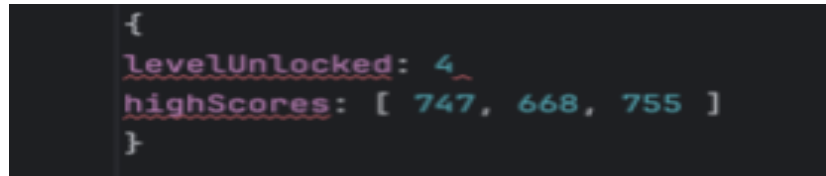


Image 4: Save.json

If a player is unable to meet the end conditions (they lose all graphite or run out of time), then the losing screen will be displayed, showing information such as score, time, and plot points collected. Here, the player cannot continue to the next level because levels must be completed sequentially to move on to the next. They can either restart or quit to the menu.



Image 5: Game over screen

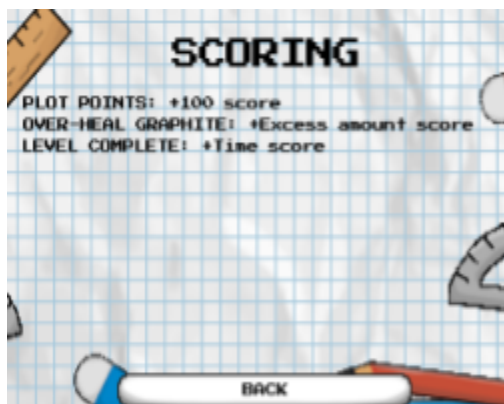


Image 6: "Scoring" menu, found within "How to Play" menu

Our game has three different methods for collecting score: plot points, excess graphite, and time. Collecting a plot point will always give you a base score of 100, regardless of your level. Excess graphite is the graphite above your max health that is collected from graphite shards. The amount varies and is determined by the amount of graphite you have at the time. Finally, we have time, which is an additional score added at the end of the game, directly coming from the time remaining when you complete the level. This encourages users to complete levels more quickly.

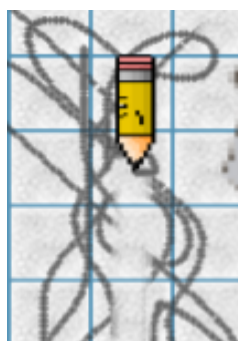


Image 8: pencil lines removed by eraser

Our game has 2 moving enemies: erasers and pencil sharpeners, who deal damage upon contact. They follow a pathfinding formula of randomly moving until the pencil is in the sight radius. The eraser's special quirk is to erase pencil lines as it walks over them, and the pencil sharpener forces you to press the space button repeatedly to escape its damage-draining trance.



Image 7: Pencil sharpener drain sequence

Link to Video:

<https://youtu.be/x5MVC6XfxyI>